

Ex 11 Profiling float

1) We have $x = \begin{pmatrix} d \\ \dot{v} \\ b \end{pmatrix}$ and we need to find $\ddot{x} = f(x, u)$ where $u = \ddot{b}$

The fundamental principle of dynamics gives us:

$$m \cdot \ddot{v} = f \text{ where } f = mg - \rho_0 g l^2 \max(0, l + \min(d, 0)) - \frac{1}{2} \rho_0 v |v| l^2 c_x$$

Knowing $v = \dot{d}$ and $m = \rho \cdot V_{\text{cube}} = (1 + \beta b) \rho_0 l^3$, we have:

$$\ddot{x} = \begin{pmatrix} \dot{d} \\ \ddot{v} \\ \ddot{b} \end{pmatrix} = \begin{pmatrix} v \\ g - \frac{g \cdot \max(0, l + \min(d, 0)) - (\frac{1}{2}) v |v| c_x}{(1 + \beta b) l} \\ u \end{pmatrix}$$

2) If we derivate $\ddot{v}$, we could find $\ddot{b}$ and make a linearization controller.

If no, we could make a sliding mode controller like $u = K \cdot \text{sign}(s(x, t))$

$$s(x, t) = (dw - d) + 2(\dot{dw} - \dot{d}) + (\ddot{dw} - \ddot{d}) \text{ for all poles equal to } -1.$$

Where dw is the target that we could easily calculate, $\dot{d} = v$ and $\ddot{d} = \ddot{v}$ from the state equation.

As u and b belong to $[-1, 1]$, the biggest $K = 1$.

4) To use a linearization feedback, we need to find u in one of d derivate.

$$\ddot{d} = \ddot{v} = \frac{d}{dt} \ddot{v}$$

$$\text{If } d > 0, \quad \ddot{v} = \frac{g \cdot l + \frac{1}{2} v |v| c_x}{(1 + \beta b) l}$$

$$\text{So } \ddot{d} = - \frac{(c_{x2}) \cdot \frac{d}{dt}(v |v|) \cdot (1 + \beta b) l - (g \cdot l + \frac{1}{2} v |v| c_x) \beta l \dot{b}}{(1 + \beta b) l^2}$$

$$\frac{d}{dt}(v |v|) = \dot{v} |v| + v \cdot \frac{d}{dt}(|v|)$$

$$\frac{d}{dt}|v| = \begin{cases} \frac{dv}{dt} & \text{if } v > 0 \\ \frac{d(-v)}{dt} & \text{if } v < 0 \end{cases} = \begin{cases} \dot{v} & \text{if } v > 0 \\ -\dot{v} & \text{if } v < 0 \end{cases} \text{ so } \frac{d}{dt}|v| = \text{sign}(v) \cdot \dot{v}$$

$$S_o, \ddot{d} = \underbrace{-\frac{C_x}{2} \frac{(\dot{v} \cdot |v| + v \cdot \text{sign}(v) \cdot \ddot{v})}{(1+\beta b)l}}_{A(x)} + \underbrace{\frac{(gl + \frac{1}{2}v^{127} \cdot C_x)\beta}{(1+\beta b)^2 \cdot l}}_{B(x)} \overset{\circ}{b} \downarrow u$$

$$\text{On a } \boxed{\ddot{d} = A(x) + B(x)u}$$

$$\text{On choisit } \boxed{u = \frac{1}{B(x)} (V_b - A(x))}$$

$$\text{avec } \boxed{V_b = (d_w - d) + 3(\dot{d}_w - \dot{d}) + 3(\ddot{d}_w - \ddot{d}) + \ddot{d}_w}$$

pour un placement de pôles en -1 .

La simulation montre que la dynamique de ce contrôle est plus lente.
L'erreur est plus lente à tendre vers 0.